Convergence of scalar ADMM with dual steps below two

Abstract

We prove convergence of the scalar alternating direction method of multipliers with an unchanged pair of primal updates and any fixed dual step $1 < \gamma < 2$. Both objective functions may be arbitrary proper closed convex functions, provided a primal-dual solution exists. The proof applies to every pair of scalar nondecreasing 1-Lipschitz maps with a fixed point of the coupled iteration. Its central estimate places each dual iterate in a convex hull determined by the preceding two dual iterates. This gives boundedness and uniform stability, and a quantitative two-step decrease proves absolute summability of the feasibility residuals. An extremal-coordinate argument shows that every compact set mapped onto itself contains a fixed point, and stability then yields convergence of the whole orbit. The result concerns scalar blocks and does not assert convergence for general vector-valued ADMM.

Keywords: Alternating direction method of multipliers; dual step length; scalar proximal maps; global convergence

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1 Introduction

Consider the equality-constrained convex problem

$$\min_{x,y\in\mathbb{R}} \; f(x) + g(y) \qquad \text{subject to } x = y. \tag{1}$$

At unit penalty, ADMM with a dual step γ takes the form

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_f(y_k - u_k), \\ y_{k+1} &= \operatorname{prox}_g(x_{k+1} + u_k), \\ u_{k+1} &= u_k + \gamma(x_{k+1} - y_{k+1}), \qquad k \geq 0. \end{aligned} \tag{2}$$

Here $\operatorname{prox}_f(t)$ minimizes $f(x) + \frac{1}{2}(x-t)^2$. Only the multiplier update is multiplied by γ; the input to the second proximal map is exactly the one displayed in (2). We study arbitrary initial values $y_0, u_0 \in \mathbb{R}$ and the entire interval $1 < \gamma < 2$.

The admissible dual step is a longstanding issue in ADMM convergence theory. Tao and Yuan [1] address Glowinski's question and prove convergence up to two for globally quadratic objective functions. Their public preprint, Theorem 4.6, provides a full-sequence statement. Chen, Sun, and Toh [2] establish a general conclusion beyond the golden ratio under an additional square-summability condition on multiplier increments; see Theorem 4.1(d) of their version 4 preprint. Chen, Li, Sun, and Toh [3] obtain convergence throughout $(0, 2)$ in a semi-proximal setting with a linear second objective block;

their Theorem 4.2 and Section 4.1 include the corresponding full-sequence conclusion under the stated rank conditions. These results permit vector variables but retain structural or summability hypotheses.

The present result permits two arbitrary nonlinear scalar proximal maps. Its assumptions neither impose quadratic structure or a linear objective block nor assume summability of the computed residual. Instead, we prove absolute summability and thereby verify the weaker square-summability premise in [2]. The proof uses order on the real line, which is not available for general vector-valued proximal maps. Gu and Yang [4] show that the usual one-step energy analysis has an obstruction beyond the golden ratio. Their rank-two construction concerns the behavior of a conventional energy, rather than a divergent scalar orbit. Here boundedness follows from a different, two-step interval estimate, and convergence follows from compactness and stability.

We first prove an operator theorem for nondecreasing 1-Lipschitz maps, then obtain (2) by translation about any primal-dual solution. Such scalar maps are the familiar one-dimensional proximal class; see, for example, Combettes and Pesquet [5], Proposition 2.4. We give directly the proximal facts needed for the convex consequence. Our statements make no claim about a last-iterate rate, a preferred solution, or the vector-valued dual-step question.

2 The scalar iteration

Let $P, Q : \mathbb{R} \to \mathbb{R}$ be nondecreasing and 1-Lipschitz. Consider

$$\begin{aligned} x_{k+1} &= P(y_k - u_k), \\ y_{k+1} &= Q(x_{k+1} + u_k), \\ u_{k+1} &= u_k + \gamma(x_{k+1} - y_{k+1}). \end{aligned} \tag{3}$$

Write $F(y_k, u_k) = (y_{k+1}, u_{k+1})$ for the state map. A state (s, v) is fixed if and only if

$$P(s - v) = s, \qquad Q(s + v) = s. \tag{4}$$

Indeed, the dual equation at a fixed state first gives $x_{k+1} = s$.

Theorem 1. *Suppose $1 < \gamma < 2$, and suppose (4) holds for at least one pair (s, v). For every $y_0, u_0 \in \mathbb{R}$, the orbit (3) converges: there is a fixed pair $(\bar{s}, \bar{v})$ such that*

$$y_k \longrightarrow \bar{s}, \qquad u_k \longrightarrow \bar{v}, \qquad x_k \longrightarrow \bar{s}.$$

Moreover, for every fixed pair (s, v) and every $k \geq 0$,

$$\max\{|y_k - s|, |u_k - v|\} \leq (2\gamma - 1) \max\{|y_0 - s|, |u_0 - v|\}. \tag{5}$$

The feasibility residuals are absolutely summable. More precisely, writing $D_v = \max\{|u_0 - v|, |u_1 - v|\}$ *for any fixed pair* (s, v)*,*

$$\sum_{k=0}^{\infty} |x_{k+1} - y_{k+1}| \le \frac{|u_1 - u_0|}{\gamma} + \frac{2D_v}{2-\gamma}. \tag{6}$$

The assertion is convergence to some fixed point. The pair used to recenter the recurrence need not be the limiting pair. The input x_0 is not used by (3), so an arbitrary separately assigned x_0 has no effect.

We use $[a, b]_{\mathrm{seg}}$ for the closed interval with endpoints a, b, regardless of their order, and co for the convex hull of finitely many real numbers. Most of the proof is carried out under the normalization

$$P(0) = Q(0) = 0. \tag{7}$$

If (s, v) satisfies (4), define

$$\widehat{P}(t) = P(t + s - v) - s, \qquad \widehat{Q}(t) = Q(t + s + v) - s. \tag{8}$$

These maps retain monotonicity and the Lipschitz constant and fix zero. The variables

$$X_{k+1} = x_{k+1} - s, \qquad Y_k = y_k - s, \qquad U_k = u_k - v$$

satisfy precisely (3) with $\widehat{P}, \widehat{Q}$. Thus normalization loses no generality.

3 A two-step bound and stability

Put

$$r = \gamma - 1 \in (0, 1), \qquad R = I - Q. \tag{9}$$

A nondecreasing 1-Lipschitz map T fixing zero obeys

$$T(t) \in [0, t]_{\mathrm{seg}} \qquad (t \in \mathbb{R}). \tag{10}$$

The map R has these same three properties. Indeed, for $a \le b$,

$$0 \le R(b) - R(a) = (b - a) - (Q(b) - Q(a)) \le b - a.$$

In particular, monotonicity of R and (10) for Q give

$$R(Q(t)) \in [0, R(t)]_{\mathrm{seg}}. \tag{11}$$

This is an interval inclusion and does not require Q and R to commute.

Define $z_k = u_k + P(y_k - u_k)$. Then (10) yields

$$z_k \in [u_k, y_k]_{\mathrm{seg}}, \tag{12}$$

and the actual recurrence becomes

$$y_{k+1} = Q(z_k), \qquad u_{k+1} = -ru_k + \gamma R(z_k). \tag{13}$$

Lemma 1. *Assume* (7). *For every* $k \geq 1$,

$$u_{k+1} \in \operatorname{co}\{u_k, -ru_k, (1-r)u_k + ru_{k-1}\}. \tag{14}$$

Consequently, with $M_0 = \max\{|u_0|, |u_1|\}$,

$$|u_k| \leq M_0 \qquad (k \geq 0). \tag{15}$$

If three successive dual values $a, b, c \in [-M, M]$ *satisfy* (14) *and* $|c| = M$, *then* $b = c$.

Proof. For $k \geq 1$, the preceding step in (13) and (11) give

$$\gamma R(y_k) \in [0, u_k + ru_{k-1}]_{\text{seg}}.$$

Also $\gamma R(u_k) \in [0, \gamma u_k]_{\text{seg}}$. By (12) and monotonicity,

$$\gamma R(z_k) \in [\gamma R(u_k), \gamma R(y_k)]_{\text{seg}} \subset \operatorname{co}\{0, \gamma u_k, u_k + ru_{k-1}\}.$$

Subtract ru_k and use $\gamma - r = 1$ to obtain (14). Each of its three generators has absolute value at most $\max\{|u_{k-1}|, |u_k|\}$, since $0 < r < 1$. Induction proves (15); no preceding step is assumed at $k = 0$.

For the equality assertion, the case $M = 0$ is immediate. When $M > 0$, multiply a, b, c by the sign of c so that $c = M$. If $b < M$, all three generators $b, -rb, (1-r)b + ra$ are strictly less than M: the second is at most $rM < M$, and the third is at most $(1-r)b + rM < M$. This contradicts membership of M in their convex hull. Thus $b = c$, also before the sign change. ∎

The hull argument uses closed intervals throughout and never divides by a coordinate or slope. Constant maps, flat intervals and zero coordinates are included.

Lemma 2. *Under* (7), *the orbit satisfies*

$$\min\{y_0, Q(-M_0)\} \leq y_k \leq \max\{y_0, Q(M_0)\} \qquad (k \geq 0). \tag{16}$$

In addition,

$$\max\{|y_k|, |u_k|\} \leq (2\gamma - 1) \max\{|y_0|, |u_0|\} \qquad (k \geq 0). \tag{17}$$

Proof. Let $a = \min\{y_0, Q(-M_0)\}$ and $b = \max\{y_0, Q(M_0)\}$. Since $Q(-M_0) \leq 0 \leq Q(M_0)$, the interval $[a, b]$ contains zero and is preserved by Q, by (10). It also contains $Q(u_k)$, by (15). If $y_k \in [a, b]$, monotonicity and (12) give

$$y_{k+1} \in [Q(u_k), Q(y_k)]_{\text{seg}} \subset [a, b].$$

Induction proves (16).

Set $D = \max\{|y_0|, |u_0|\}$. Equation (12) gives $|z_0| \leq D$; hence (13) and (10) for R imply

$$|u_1| \leq r|u_0| + \gamma|R(z_0)| \leq (r + \gamma)D = (2\gamma - 1)D.$$

Thus $M_0 \leq (2\gamma - 1)D$. The primal interval bound and $|Q(\pm M_0)| \leq M_0$ prove (17). ∎

Applying Lemma 2 after the translation (8) proves (5) about every fixed point. In particular, every fixed point is Lyapunov stable, with a stability constant independent of the initial state and the number of iterations.

3.1 Finite variation of the dual sequence

The hull inclusion also yields a quantitative decrease over two steps. The following lemma concerns arbitrary scalar sequences satisfying that inclusion, so its conclusion uses no prior convergence assertion.

Lemma 3. *Let $0 < r < 1$, and suppose a real sequence satisfies* (14) *for every $k \geq 1$. Put*

$$M_k = \max\{|u_{k-1}|, |u_k|\}, \qquad \eta = \frac{1-r}{1+r}.$$

Then, for every $k \geq 1$,

$$M_{k+2} \leq M_k - \eta|u_{k+1} - u_k|. \tag{18}$$

Consequently,

$$\sum_{k=1}^{\infty} |u_{k+1} - u_k| \leq \frac{M_1 + M_2}{\eta} \leq \frac{2M_1}{\eta}. \tag{19}$$

Proof. For real $a, b \in [-M, M]$ and $0 \leq t \leq 1$, one has

$$M - |(1-t)b + ta| \geq \min\{t, 1-t\}|a - b|. \tag{20}$$

Indeed, both $M - ((1-t)b + ta)$ and $M + ((1-t)b + ta)$ are bounded below by the right side: write them as weighted sums of $M \mp b$ and $M \mp a$, and use $2M \mp (a+b) \geq |a-b|$.

Suppose $c \in \mathrm{co}\{b, -rb, (1-r)b + ra\}$. We claim

$$|c| + \eta|c - b| \leq M, \tag{21}$$

$$|(1-r)c + rb| + \eta|c - b| \leq M. \tag{22}$$

Each left side is convex as a function of c, so it suffices to check the three generators. At $c = b$ both claims reduce to $|b| \leq M$. At $c = -rb$, their left sides are respectively

$$[r + \eta(1+r)]|b| = |b|, \qquad [r^2 + \eta(1+r)]|b| = (r^2 + 1 - r)|b| \leq M.$$

At $c = (1-r)b + ra$, estimate (20) with $t = r$ gives (21), since $\min\{r, 1-r\} \geq \eta r$ and $|c - b| = r|a - b|$. For (22), note that

$$(1-r)c + rb = (1 - r(1-r))b + r(1-r)a.$$

Apply (20) with $t = r(1-r) \leq 1/4$. Its right side is $r(1-r)|a-b| \geq \eta|c-b|$. This proves both claims, including $M = 0$.

If $d \in \mathrm{co}\{c, -rc, (1-r)c + rb\}$, equations (21)–(22) bound the absolute value of each generator by $M - \eta|c - b|$. They also bound $|c|$ by this quantity. Therefore

$$\max\{|c|, |d|\} \leq M - \eta|c - b|.$$

Use $a = u_{k-1}$, $b = u_k$, $c = u_{k+1}$ and $d = u_{k+2}$ to obtain (18). Summing it for $k = 1, \dots, n$ gives

$$\eta \sum_{k=1}^{n} |u_{k+1} - u_k| \le M_1 + M_2 - M_{n+1} - M_{n+2} \le M_1 + M_2.$$

The partial sums are nonnegative and increasing. Letting $n \to \infty$, and using the one-step bound $M_2 \le M_1$, proves (19). ∎

Apply Lemma 3 to $U_k = u_k - v$ after translation about a fixed pair. Since $r = \gamma - 1$, one has $\eta = (2 - \gamma)/\gamma$ and $M_1 = D_v$. The identity

$$u_{k+1} - u_k = \gamma(x_{k+1} - y_{k+1})$$

then proves (6), with the $k = 0$ term treated separately. Thus the dual sequence has finite total variation. In the convex ADMM setting this also supplies the square-summability condition of [2, Theorem 4.1(d)], giving another route to full convergence. We retain the self-contained compactness argument below. No finite-length assertion for the full primal-dual path is needed.

4 Compact invariant sets and convergence

Boundedness alone does not give convergence. The equality case of Lemma 1 supplies the additional information needed on an omega-limit set.

Lemma 4. *Under* (7)*, let* $K \subset \mathbb{R}^2$ *be nonempty and compact with* $F(K) = K$*. Then* K *contains a fixed point of* F*.*

Proof. Let

$$M = \max\{|u| : (y, u) \in K\}.$$

Choose v with $|v| = M$ such that the compact slice

$$K_v = \{y : (y, v) \in K\}$$

is nonempty. We first show that every point $(y, v) \in K$ has a predecessor in the same slice. Since $F(K) = K$, there exist $w_0, w_1 \in K$ with

$$F(w_0) = w_1, \qquad F(w_1) = (y, v).$$

Apply (14) to this length-two orbit. All three dual coordinates lie in $[-M, M]$, and the last is v. The equality assertion of Lemma 1 forces the dual coordinate of w_1 to be v. This includes $M = 0$. Only two preimages are needed.

Write such a predecessor as (y', v). Its unchanged dual coordinate implies that its primal output x^+ equals y. Thus

$$y = P(y' - v), \qquad Q(y + v) = y. \tag{23}$$

In particular, the second identity holds for every $y \in K_v$. Let $\ell = \min K_v$ and $h = \max K_v$. A predecessor $\ell' \in K_v$ of ℓ has $\ell' \geq \ell$, so monotonicity gives

$$P(\ell - v) \leq P(\ell' - v) = \ell. \tag{24}$$

A predecessor $h' \in K_v$ of h has $h' \leq h$, yielding

$$P(h - v) \geq P(h' - v) = h. \tag{25}$$

When $\ell < h$, the Lipschitz property gives

$$h - \ell \leq P(h - v) - P(\ell - v) \leq h - \ell.$$

Both (24) and (25) must therefore be equalities. When $\ell = h$, these two inequalities directly give the same conclusion. In particular,

$$P(\ell - v) = \ell, \qquad Q(\ell + v) = \ell,$$

where the second identity follows from (23). Hence $(\ell, v) \in K$ is fixed. ∎

The hypothesis is the onto relation $F(K) = K$, rather than forward invariance alone. The slice K_v need not itself be forward invariant. The predecessor argument uses neither invertibility of F nor strict monotonicity of either scalar map.

Proof of Theorem 1. First assume (7). Lemma 2 bounds the state orbit $w_k = (y_k, u_k)$, and F is continuous. Its omega-limit set

$$\Omega = \bigcap_{m \geq 0} \overline{\{w_k : k \geq m\}}$$

is nonempty and compact. Continuity gives $F(\Omega) \subset \Omega$. For the reverse inclusion, let $w_{k_j} \to w \in \Omega$ with $k_j \to \infty$. A subsequence of the bounded predecessors $w_{k_j - 1}$ converges to some $a \in \Omega$, and continuity gives $F(a) = w$. Thus $F(\Omega) = \Omega$.

By Lemma 4, there is a fixed point $w_* = (\bar{s}, \bar{v}) \in \Omega$. The stability estimate (5), already proved using (8), now turns this fixed cluster point into the limit of the whole orbit. Given $\varepsilon > 0$, choose an index n from a subsequence converging to w_* such that

$$\|w_n - w_*\|_\infty < \frac{\varepsilon}{2\gamma - 1}.$$

The iteration is autonomous. Restarting the estimate at n gives, for every $j \geq 0$,

$$\|w_{n+j} - w_*\|_\infty \leq (2\gamma - 1)\|w_n - w_*\|_\infty < \varepsilon.$$

Therefore $w_k \to w_*$. Continuity of P and fixedness give

$$x_{k+1} = P(y_k - u_k) \longrightarrow P(\bar{s} - \bar{v}) = \bar{s}.$$

For general P, Q with a fixed pair, apply the normalized argument to (8) and translate the conclusion back. The stability and residual-sum bounds were already proved in their original coordinates. ∎

5 Convex scalar ADMM

We now make the convex consequence explicit, including extended-valued objectives and arbitrary initialization. For a proper closed convex function $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$, write

$$\partial f(s) = \{a \in \mathbb{R} : f(t) \geq f(s) + a(t - s) \text{ for every } t \in \mathbb{R}\}.$$

The subdifferential is empty when $f(s) = +\infty$.

Corollary 5.1. *Let $f, g : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ be proper closed convex functions. Suppose that a primal-dual solution exists, meaning that some $(s, u^*) \in \mathbb{R}^2$ satisfies*

$$-u^* \in \partial f(s), \qquad u^* \in \partial g(s). \tag{26}$$

For every fixed $1 < \gamma < 2$ and every $y_0, u_0 \in \mathbb{R}$, all subproblems in (2) have unique solutions, and there is a pair $(\bar{s}, \bar{u})$ satisfying (26) such that

$$x_k \longrightarrow \bar{s}, \qquad y_k \longrightarrow \bar{s}, \qquad u_k \longrightarrow \bar{u}.$$

In addition, the bound (6) holds for every KKT pair (s, v). In particular, $\bar{s}$ minimizes $f + g$. No uniqueness of the primal or dual solution is assumed.

Proof. A proper closed convex function on $\mathbb{R}$ has an affine lower bound. For a singleton effective domain this is immediate; otherwise a supporting slope at an interior point of its effective domain gives such a bound by the usual ordering of convex secant slopes. Hence $f(x) + \frac{1}{2}(x - t)^2$ is coercive and lower semicontinuous for every $t \in \mathbb{R}$, and it attains its minimum. Strict convexity of the quadratic makes the minimizer unique.

The proximal optimality condition is

$$p = \operatorname{prox}_f(t) \quad \Longleftrightarrow \quad t - p \in \partial f(p). \tag{27}$$

For completeness, if p minimizes the proximal objective, compare its value with the value at $p + \theta(z - p)$, use convexity of f, divide by $\theta > 0$, and let $\theta \downarrow 0$. This gives $f(z) \geq f(p) + (t - p)(z - p)$ for every z with finite function value, and the inequality is automatic otherwise. Conversely, this subgradient inequality bounds the proximal objective difference at z below by $\frac{1}{2}(z - p)^2$.

If $p_i = \operatorname{prox}_f(t_i)$, monotonicity of the subdifferential, obtained by adding its two defining inequalities, gives

$$(p_1 - p_2)(t_1 - t_2) \geq (p_1 - p_2)^2.$$

For $t_1 > t_2$, this implies $0 \leq p_1 - p_2 \leq t_1 - t_2$. Thus prox_f is nondecreasing and 1-Lipschitz. The same holds for prox_g.

By (27), the assumed pair in (26) satisfies

$$\operatorname{prox}_f(s-u^*)=s, \qquad \operatorname{prox}_g(s+u^*)=s.$$

Theorem 1, with $P=\operatorname{prox}_f$ and $Q=\operatorname{prox}_g$, proves convergence to a fixed pair $(\bar{s},\bar{u})$. Applying (27) at this limiting pair yields

$$-\bar{u}\in\partial f(\bar{s}), \qquad \bar{u}\in\partial g(\bar{s}).$$

Adding their subgradient inequalities shows that $\bar{s}$ minimizes $f+g$. ∎

6 Scope and limiting cases

The proof controls every orbit of the stated scalar iteration and proves absolute residual summability. The scalar hull inclusion (14) supplies both a two-step decrease and uniform stability; the extremal-slice argument in Lemma 4 supplies a fixed cluster point. The passage from a stable fixed cluster point to full convergence is a standard stability argument.

The constants and degenerate cases can be checked directly. With $P=I$, $Q=0$, $y_0=d$ and $u_0=-d$ for $d>0$, the first update gives

$$y_1=0, \qquad u_1=(2\gamma-1)d.$$

Since $(0,0)$ is fixed, the factor in (5) cannot be reduced uniformly over this operator class. These maps are proximal maps of the zero function and the indicator of $\{0\}$, respectively. At the excluded endpoint $\gamma=2$, the same maps with $y_0=0, u_0=1$ give $u_k=(-1)^k$, so convergence fails. For $\gamma>2$ the magnitude grows geometrically. These are elementary linear endpoint examples; the nonlinear content is convergence for arbitrary scalar proximal maps below two.

Nonunique fixed points are also included. If $P=Q=0$, then $F(y,u)=(0,u)$: the dual variable stays at its initial value, and every $(0,u)$ is fixed. If $P=Q=I$, then $y_k=y_0$ and $u_k=(1-\gamma)^k u_0$, so the limit is $(y_0,0)$. Accordingly, the theorem gives convergence to some solution and makes no projection or minimum-norm selection claim.

The iteration (2) changes only the dual step and keeps the penalty fixed. A relaxed ADMM that also replaces the first primal output before the second subproblem is a different recurrence. No variable-penalty, noisy, inexact-subproblem, or general vector-valued conclusion is inferred here.

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